

ARYA QUADRUPED ROBOT PROTOTYPE

BILL OF MATERIALS

262.77 x 213.22 x 186.23 mm | Nylon 101 | CAD Mass: 798.37 g | 8 x MG996R Servos

SECTION 1 - ELECTRONICS

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
1	Raspberry Pi 4 Model B 4 GB	Main computer, ROS, Vision	1	4500	4500
2	PCA9685 16-Channel Servo Driver	PWM servo driver, I2C	1	250	250
3	IMU Module BNO055 or MPU9250	9-DOF, orientation and balance	1	400	400
4	CSI Camera Module	Raspberry Pi Camera v2	1	600	600
5	LiPo Battery 3S 11.1 V 2200 mAh	Main power source	1	1200	1200
6	DC Power Switch	ON/OFF toggle	1	50	50

7	Buck Converter 11.1 V to 5 V 3 A	Powers Raspberry Pi and electronics	1	150	150
8	Power Distribution Board	Custom PCB or prefab	1	200	200
9	Micro SD Card 32 to 64 GB	Operating system storage	1	350	350
Section 1 Total					INR 7,700

SECTION 2 - ACTUATORS

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
10	MG996R Servo Motor - Hip	Metal gear, 11 kg.cm at 6 V	4	350	1400
11	MG996R Servo Motor - Knee	Metal gear, 11 kg.cm at 6 V	4	350	1400
12	Servo Horn Aluminum or Plastic	Diameter 24 mm, M3 centre bore	8	40	320
Section 2 Total					INR 3,120

SECTION 3 - MECHANICAL PARTS (Nylon 101 FDM Printed)

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
13	Upper Chassis Plate	Nylon 101, 262.77 x 213.22 x 6 mm	1	180	180
14	Lower Chassis Plate	Nylon 101, 262.77 x 213.22 x 6 mm	1	160	160
15	Electronics Plate	Nylon 101, 180 x 100 x 4 mm	1	80	80
16	Hip Servo Bracket	Nylon 101, 40 x 35 x 25 mm	4	60	240
17	Knee Servo Bracket	Nylon 101, 35 x 30 x 25 mm	4	50	200
18	Thigh Link	Nylon 101, 80 x 15 x 8 mm	8	30	240
19	Shin Link	Nylon 101, 70 x 15 x 8 mm	8	25	200
20	Foot Connector	Nylon 101, 20 x 12 x 12 mm	4	20	80

21	Rubber Foot Pad	TPU 95A, Diameter 25 x 6 mm	4	20	80
22	Battery Tray	Nylon 101, 115 x 45 x 30 mm	1	70	70
23	Camera Mount	Nylon 101, 40 x 35 x 25 mm	1	40	40
24	IMU Holder	Nylon 101, 25 x 25 x 10 mm	1	20	20

Prices shown are estimated filament and print cost only. Actual cost depends on your printer setup and filament brand.

Section 3 Total	INR 1,590
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SECTION 4 - FASTENERS AND HARDWARE

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
25	Socket Head Cap Screw M3 x 8 mm	ISO 4762, Stainless Steel A2	50	2	100
26	Socket Head Cap Screw M3 x 12 mm	ISO 4762, Stainless Steel A2	20	3	60
27	Socket Head Cap Screw M3 x 16 mm	ISO 4762, Stainless Steel A2	20	3	60

28	Hex Nut M3	ISO 4032, Stainless Steel	52	1	52
29	Flat Washer M3	ISO 7089, Stainless Steel	52	1	52
30	Spring Washer M3	DIN 127, Stainless Steel	52	1	52
31	Brass Heat-Set Insert M3	M3 x 5 mm length x 4 mm OD	24	5	120
32	Clevis Pin Headed ISO 2341	Diameter 6 mm x 30 mm, mild steel	8	15	120
Section 4 Total					INR 616

SECTION 5 - WIRING AND CONNECTORS

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
33	22 AWG Silicone Wire Red	2 m roll	1	80	80
34	22 AWG Silicone Wire Black	2 m roll	1	80	80

35	26 AWG Servo Signal Wire	5 m roll	1	100	100
36	Female Dupont Connectors 2.54 mm	Pack of 30	1	60	60
37	Male Dupont Connectors 2.54 mm	Pack of 30	1	60	60
38	JST-XH Battery Connector	2S / 3S balance plug	1	30	30
39	XT60 Connector Male and Female	30 A rated, pair	2	40	80
40	Heat Shrink Tubing	Assorted 2 to 6 mm	1	60	60
41	Nylon Cable Ties 2.5 mm	100 mm length, pack of 30	1	30	30
42	Braided Cable Sleeve	Diameter 6 to 10 mm, 1 m	1	80	80
Section 5 Total					INR 660

SECTION 6 - BEARINGS

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
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43	Ball Bearing MR106ZZ	6 x 10 x 3 mm, shielded	8	30	240
Section 6 Total					INR 240

SECTION 7 - FILAMENT (3D Printing Material)

Sr.	Component	Specification	Qty	Unit Price (INR)	Total Price (INR)
44	Nylon 101 Filament PA12	1 kg spool, 1.75 mm diameter	1	2500	2500
45	TPU 95A Filament	500 g spool, 1.75 mm diameter	1	900	900

Nylon 101 requires enclosed printer, 250 to 260 degrees C nozzle, 70 to 80 degrees C bed, and dry filament. Dry at 80 degrees C for 6 to 8 hours before printing.

Section 7 Total	INR 3,400
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GRAND TOTAL (All Sections Combined)	INR 17,326
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Approximate USD equivalent: USD 208 | All prices are estimates based on Indian market (Robu.in, Amazon India, local hardware). Actual prices may vary.

PROTOTYPE SPECIFICATIONS SUMMARY

Parameter	Value
Overall Dimensions (L x W x H)	262.77 x 213.22 x 186.23 mm
CAD Structural Mass	798.37 g (SolidWorks confirmed)
Estimated All-Up Mass	Approx 1.2 kg (with all electronics)
Structural Material	Nylon 101 PA12 (FDM 3D Printed)
Foot Pad Material	Rubber or TPU 95A
Servo Motors	8 x MG996R Metal Gear (4 hip + 4 knee)
Servo Torque at 6 V	11 kg.cm = 1.08 N.m per servo
Hip Torque Required (design load)	0.21 N.m — Safety margin 5.1x — SAFE
Knee Torque Required (design load)	0.18 N.m — Safety margin 6.0x — SAFE
Controller	Raspberry Pi 4 Model B 4 GB

Servo Driver	PCA9685 16-Channel PWM via I2C
IMU	BNO055 or MPU9250 9-DOF
Camera	Raspberry Pi CSI Camera Module
Battery	3S LiPo 11.1 V, 2200 to 3000 mAh
Total DOF	8 (2 per leg: hip pitch and knee pitch)
Gait Type	Crawl gait, 3-leg support model
Estimated Total Project Cost	INR 17,326 (approx USD 208)